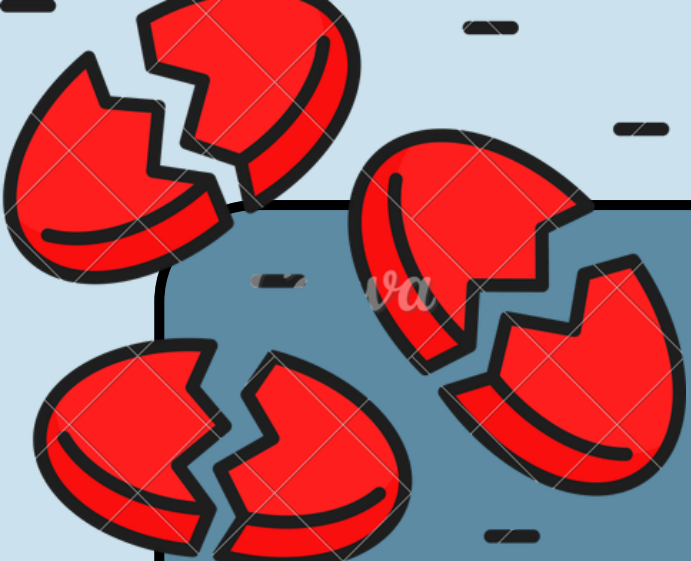
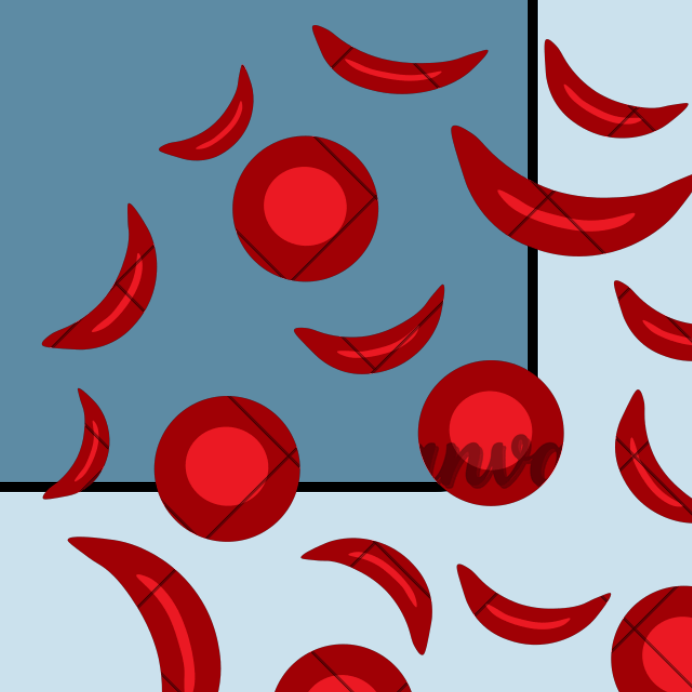




SICKLE CELL ANEMIA

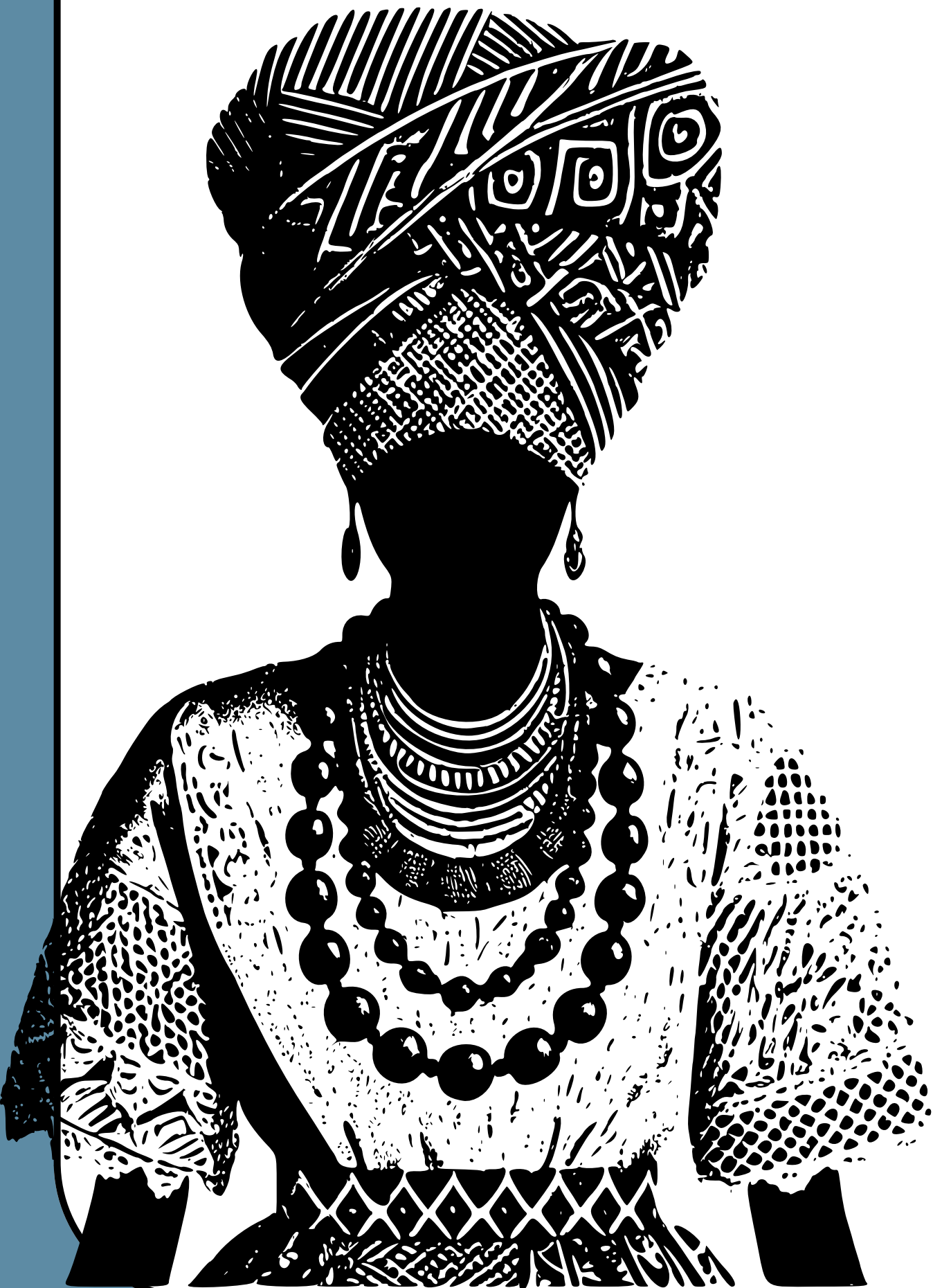
The Silent Killer of Africa



**WHERE DOES SICKLE CELL
ANEMIA FIRST BEGAIN???**



THOUSANDS OF YEARS AGO!!!



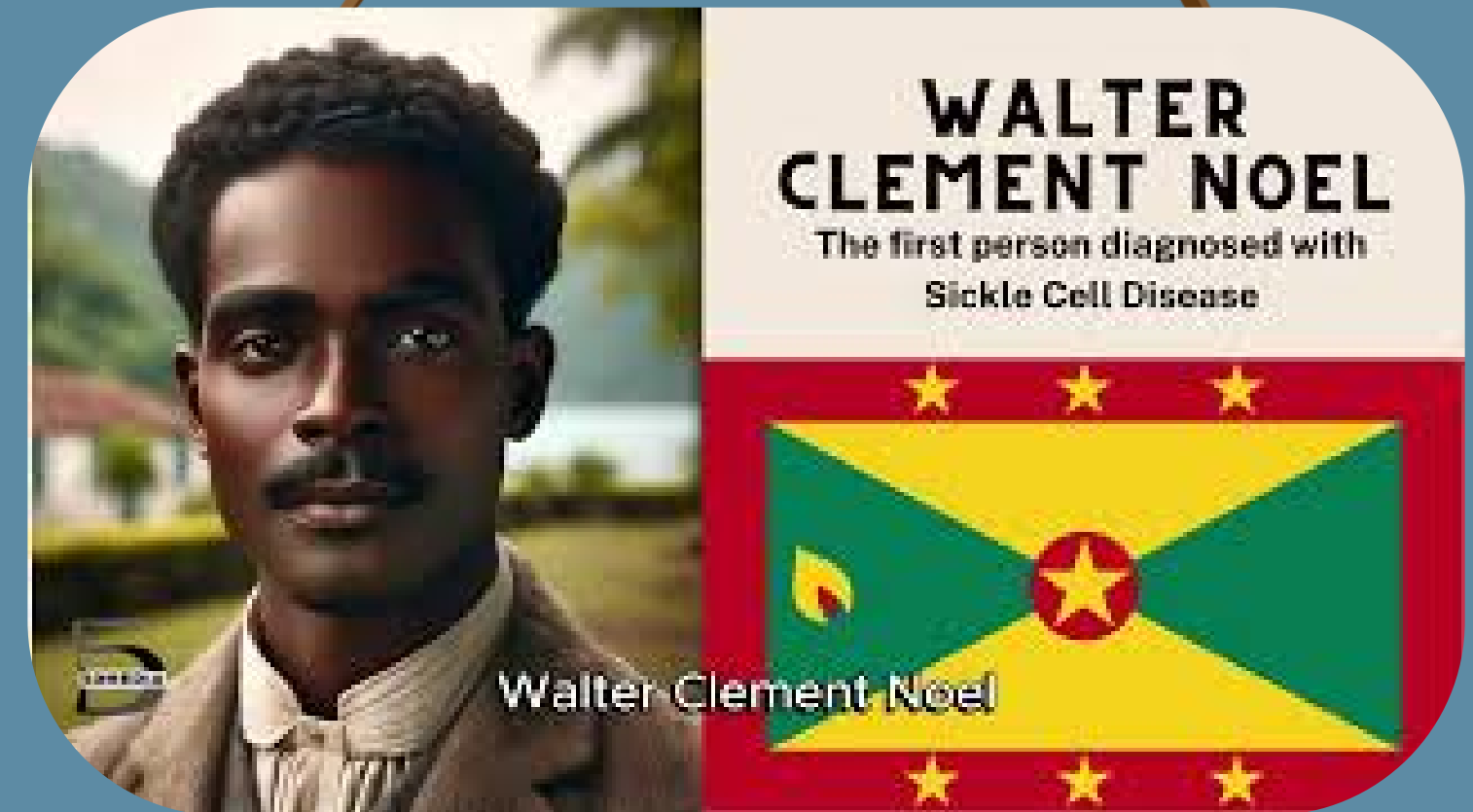
OGABANJE!!!!!!

WHEN DID FIRST SICKLE CELL CASE OCCUR??

IN USA AT 1904-1910

CHICAGO PRESBYTERIAN HOSPITAL

DR. JAMES B. HERRICK & ERNEST E. IRONS



Peculiar elongated and sickle-shaped red blood corpuscles in a case of severe anemia. 1910.

[J B Herrick](#)

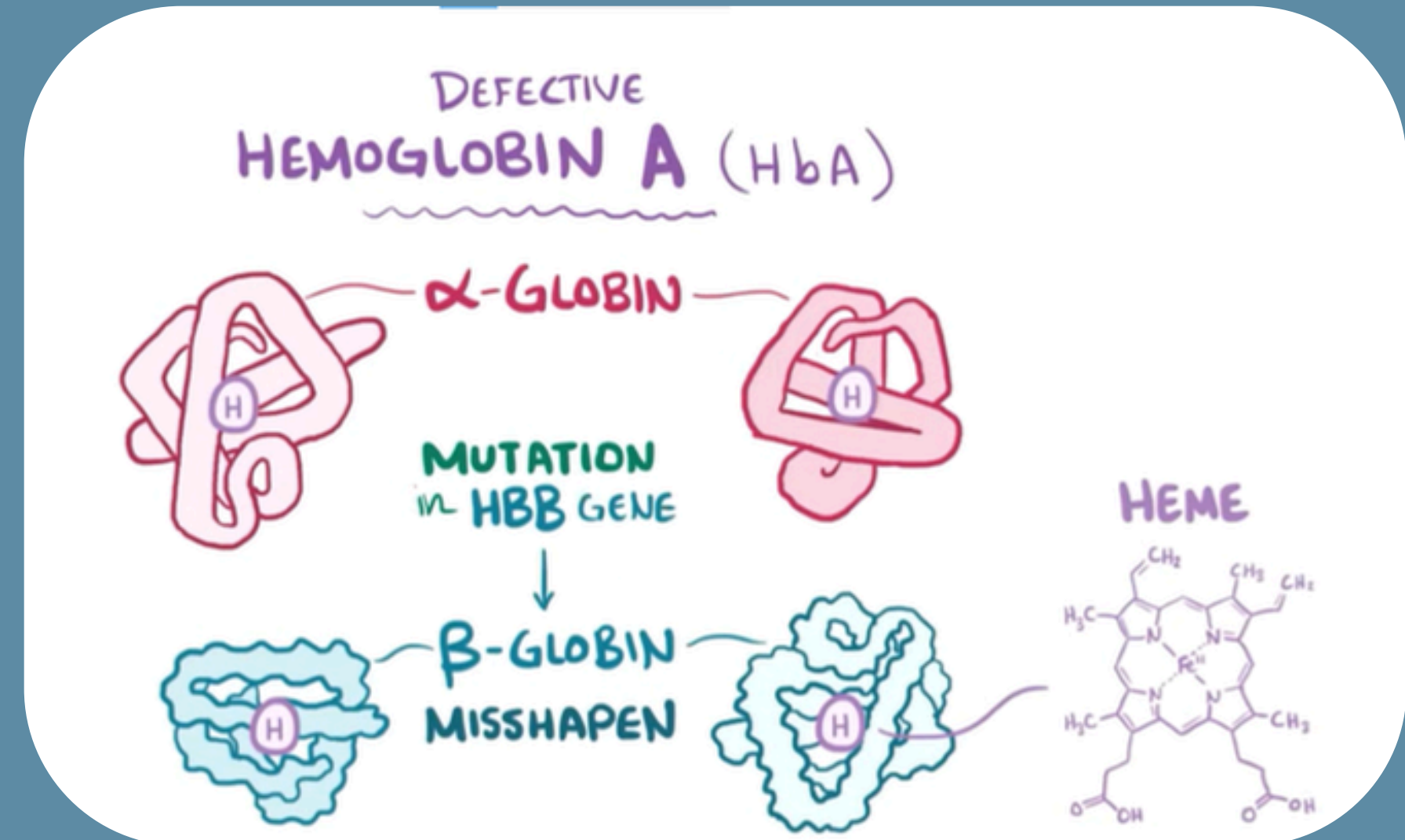
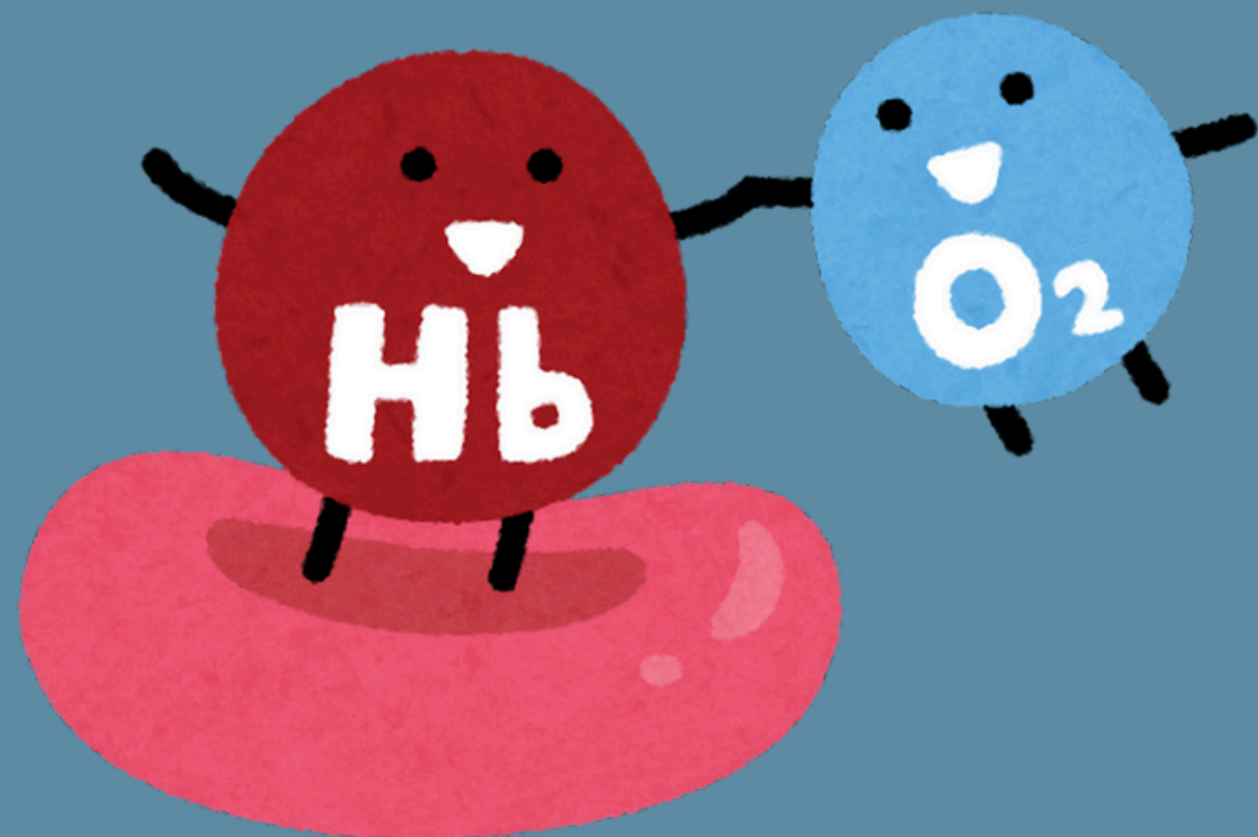
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PMCID: PMC2588723 PMID: [11501714](#)

PATHOPHYSIOLOGY

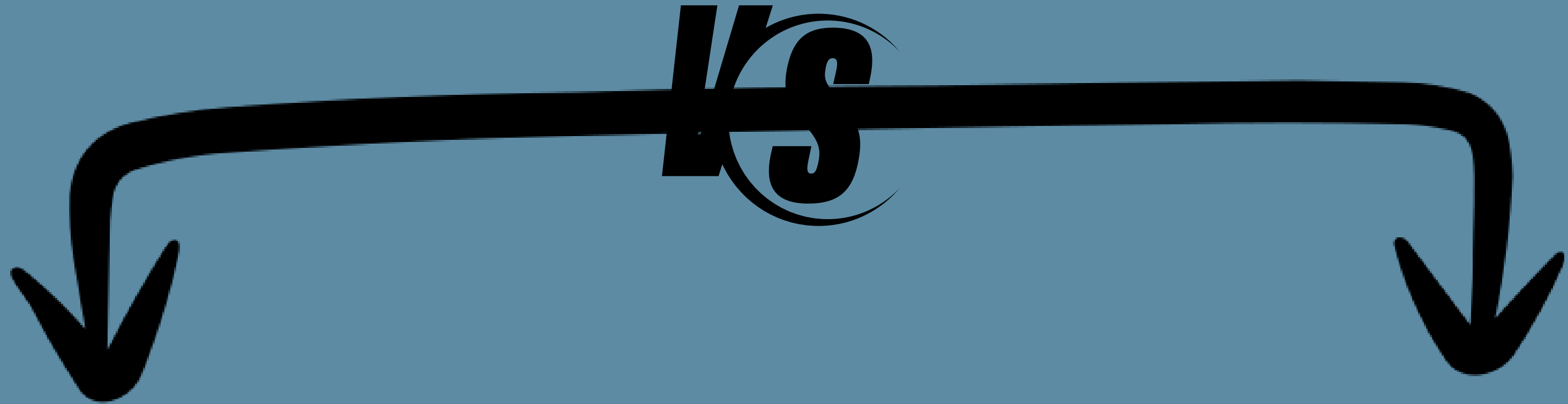


BETWEEN 1927 AND 1956, SCIENTISTS INVESTIGATED THE MECHANISM OF THE DISEASE AND IDENTIFIED THE LOCATION OF THE GENETIC MUTATION RESPONSIBLE FOR IT.



**BETA-GLOBIN GENE
GLUTAMIC ACID ----> VALINE
DECREASING OF NITRIC OXIDE IN BLOOD.**

MODELS OF ORIGIN



MULTICENTRIC MODEL

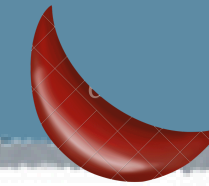
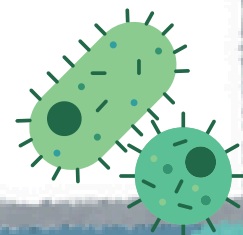
SICKLE CELL GENE ORIGINATED FROM FOUR SEPARATE MUTATIONS: THREE IN AFRICA AND ONE IN SAUDI ARABIA OR CENTRAL INDIA, ABOUT 70,000–150,000 YEARS AGO.

UNICENTRIC MODEL

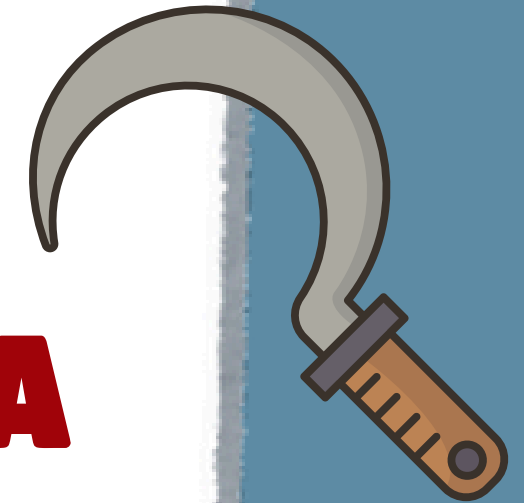
**SICKLE CELL MUTATION LIKELY AROSE IN ONE PERSON IN WESTERN AFRICA (MODERN CAMEROON) ABOUT 7,300 YEARS AGO.
ALL FIVE HAPLOTYPES IN CAMEROON AND EGYPT SUGGEST ITS LONG PRESENCE IN THESE REGIONS.**



MALARIA



**SICKLE
CELL
ANEMIA**



TREATMENTS OF SCD



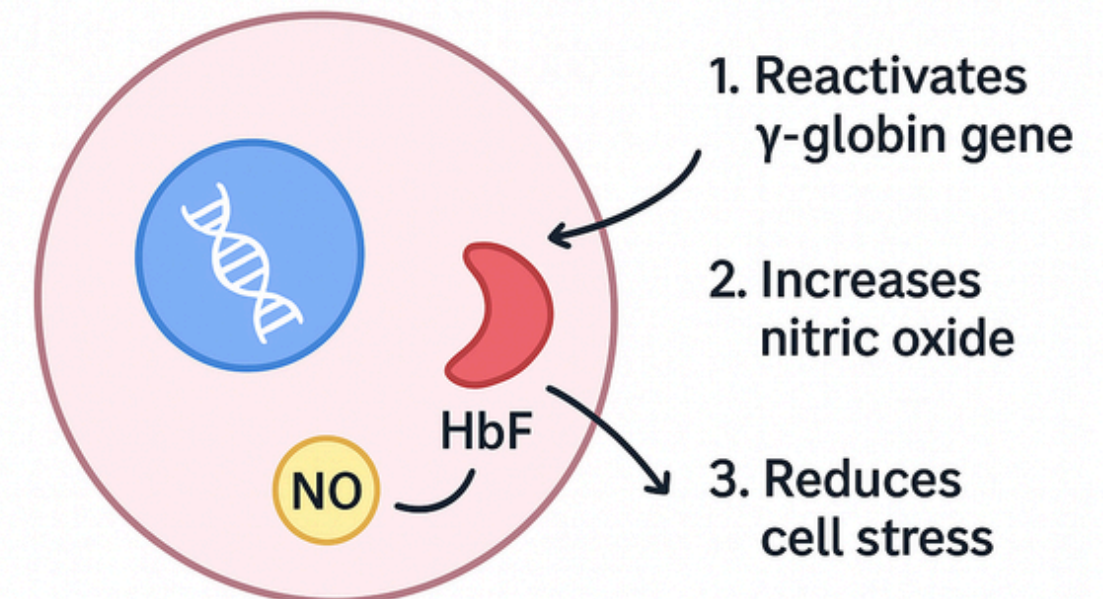
Pharmacologic Therapy

HYDROXYUREA:

- ◆ INCREASE IN FETAL HEMOGLOBIN (HbF) IN RBCS.
- ◆ HbF DOES NOT POLYMERIZE LIKE SICKLE HEMOGLOBIN (HbS),
- ◆ PREVENTS SICKLING OF RED BLOOD CELLS.
- ◆ REDUCES HEMOLYSIS (CELL BREAKDOWN).
- ◆ IMPROVES OXYGEN DELIVERY.
- ◆ DECREASES PAINFUL VASO-OCCLUSIVE CRISES (BLOCKAGES IN BLOOD VESSELS).

HbF?!!

HYDROXYUREA REACTIVATES HbF



Pharmacologic Therapy

L-GLUTAMINE

- ◆ **REDUCES PAIN CRISES**
- ◆ **IMPROVES RED BLOOD CELL STABILITY**

VOXELOTOR (GBT440)

- ◆ **INCREASES HEMOGLOBIN LEVELS**
- ◆ **PREVENTS SICKLING**
- ◆ **REDUCES ANEMIA**

CRIZANLIZUMAB



Blood-Based Therapy

BLOOD TRANSFUSION

- ◆ PREVENTS STROKE IN CHILDREN
- ◆ IMPROVES OXYGEN DELIVERY

RISKS

- ⚠ IRON OVERLOAD
- ⚠ ALLOIMMUNIZATION.



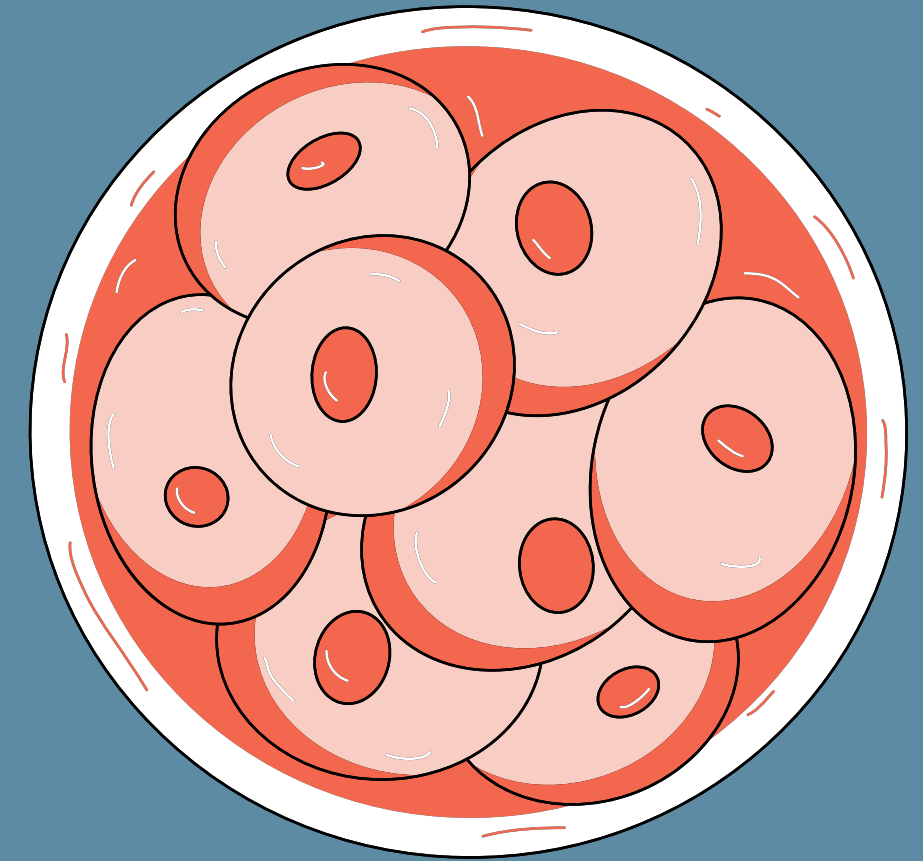
Stem cell transplant

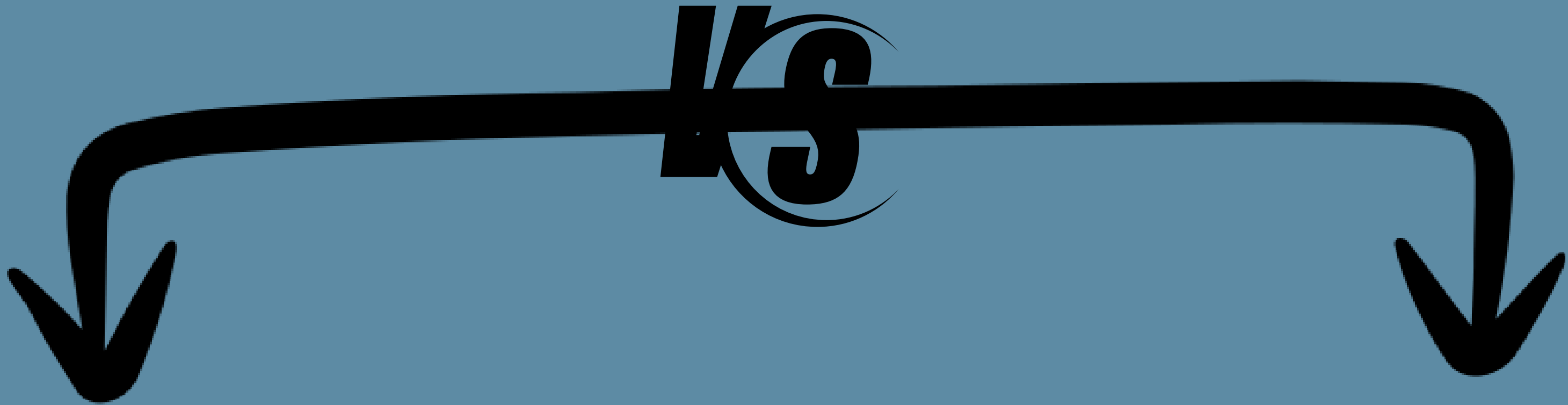
HEMATOPOIETIC STEM CELL TRANSPLANT (HSCT)

- ◆ **REPLACES THE PATIENT'S DEFECTIVE BONE MARROW WITH HEALTHY DONOR STEM CELLS**
- ◆ **ALLOWS THE BODY TO PRODUCE NORMAL RED BLOOD CELLS.**

RISKS

- ⚠ **IMMUNE REJECTION.**
- ⚠ **INFERTILITY**





~~Ks~~

STEM CELL GENE ADDITION THERAPY

PATIENT'S OWN STEM CELLS ARE COLLECTED, MODIFIED WITH A HEALTHY HEMOGLOBIN-PRODUCING GENE, AND THEN RETURNED THROUGH AN AUTOLOGOUS TRANSPLANT.

GENE EDITING THERAPY

DIRECTLY CORRECTS THE DEFECTIVE SICKLE GENE IN THE PATIENT'S STEM CELLS. THE EDITED CELLS ARE THEN INFUSED BACK INTO THE BLOODSTREAM TO RESTORE NORMAL RED BLOOD CELL PRODUCTION.

REFERENCES



- <https://n9.cl/sjhn1>
- <https://n9.cl/ok700>
- <https://n9.cl/uz227k>
- <https://n9.cl/zvwja>
- <https://n9.cl/katyh>



